

## 1.4 Linear Combinations and Span

In the previous section, we studied subspaces. A natural question arises: given a set of vectors, how can we describe the “smallest” subspace that contains them? This leads us to the concept of linear combinations and the span.

### 1.4.1 Linear Combinations

#### Definition 1.4.1 — Linear Combination

Let  $V$  be a vector space over  $F$ ,  $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$  be a subset of  $V$ . Then the expression

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_m\mathbf{v}_m$$

is called a **linear combination** of  $S$ . The scalars  $a_i$  are called the **coefficients** of the linear combination.

#### Example. (Examples of Linear Combinations)

Below are some examples of linear combinations in different vector spaces:

1. The vector  $(2, 1, 0)$  in  $\mathbb{R}^3$  is a linear combination of  $S = \{(1, 2, 3), (4, 5, 6)\}$  because

$$(2, 1, 0) = -2 \cdot (1, 2, 3) + 1 \cdot (4, 5, 6).$$

2. The polynomial  $f(x) = x^3 + 2x + 1$  in  $\mathbb{R}[x]$  is a linear combination of  $S = \{1, x, x^2, x^3\}$  because

$$f(x) = 1 \cdot 1 + 2 \cdot x + 0 \cdot x^2 + 1 \cdot x^3.$$

### 1.4.2 Span and Spanning Sets

#### Definition 1.4.2 — Span

Let  $S$  be a subset of a vector space  $V$  over  $F$ . The **span** of  $S$ , denoted by  $\text{span}_F(S)$ , is the set of all linear combinations of finite subsets of elements of  $S$ .

If  $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ , then  $\text{span}_F(S) = \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n : a_i \in F\}$ . By convention,  $\text{span}_F(\emptyset) = \{\mathbf{0}\}$ .

*Remark.* When the underlying field is clear, we abuse the notation of  $\text{span}_F$  and write it as  $\text{span}$ .

#### Theorem 1.4.1 — Spanning Set as Subspace